3D-R2N2: A Unified Approach for Single and Multi-view 3D Object Reconstruction

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Abstract. Inspired by the recent success of methods that employ shape priors to achieve robust 3D reconstructions, we propose a novel recurrent neural network architecture that we call the 3D Recurrent Reconstruction Neural Network (3D-R2N2). The network learns a mapping from images of objects to their underlying 3D shapes from a large collection of synthetic data [13]. Our network takes in one or more images of an object instance from arbitrary viewpoints and outputs a reconstruction of the object in the form of a 3D occupancy grid. Unlike most of the previous works, our network does not require any image annotations or object class labels for training or testing. Our extensive experimental analysis shows that our reconstruction framework (i) outperforms the state-of-the-art methods for single view reconstruction, and (ii) enables the 3D reconstruction of objects in situations when traditional SFM/SLAM methods fail (because of lack of texture and/or wide baseline).

Keywords: Multi-view · Reconstruction · Recurrent neural network

1 Introduction

Rapid and automatic 3D object prototyping has become a game-changing innovation in many applications related to e-commerce, visualization, and architecture, to name a few. This trend has been boosted now that 3D printing is a democratized technology and 3D acquisition methods are accurate and efficient [15]. Moreover, the trend is also coupled with the diffusion of large scale repositories of 3D object models such as ShapeNet [13].

Most of the state-of-the-art methods for 3D object reconstruction, however, are subject to a number of restrictions. Some restrictions are that: (i) objects must be observed from a dense number of views; or equivalently, views must have a relatively small baseline. This is an issue when users wish to reconstruct

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